AI Shopping Agent – Intelligent E-Commerce Recommendation System

Project Id: 25-26J-351

Project Proposal Report

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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

..... 27/08/2025

Signature of the supervisor: Date:

ABSTRACT

Many people shop online these days because it's easy and quick [1],[5]. You can buy almost anything from your house, including clothes, toys, and food. It can be difficult to find the precise item you're looking for, though, because there are so many products available. There are times when you search through numerous pages for a long time without finding the best product.

The goal of this project is to develop an AI shopping agent that facilitates better online shopping. This intelligent assistant makes use of computer programs known as natural language processing (NLP) and artificial intelligence (AI). The system gains a better understanding of the shopper's true desires thanks to these tools [7],[8]. It learns from the user's previous looks, purchases, and searches [13],[14]. Additionally, when conversing with a friend, it can understand spoken or written requests or questions in common language.

The AI shopping agent can therefore suggest goods that fit the user's requirements and price range. The system keeps learning and getting better as more people use it. As a result, customers can shop faster and with less confusion.

This tool can help businesses increase sales by showcasing products that customers like [16],[18]. Spending less time searching and more time enjoying their purchases benefits customers. Online shopping is made more convenient and personalized by the AI Shopping Agent.

To put it briefly, this project develops a clever shopping assistant that makes online shopping more enjoyable, saves time, and assists users in finding the right products quickly.

Keywords: *AI Shopping Agent, Online Shopping, Personalized Recommendations, NLP, AI, Product Suggestions, User Preferences, Smart E-Commerce, Shopping Assistant, Customer Experience.*

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LIST OF ABBREVIATIONS

- **AI** – Artificial Intelligence
- **NLP** – Natural Language Processing
- **UI** – User Interface
- **API** – Application Programming Interface
- **DB** – Database

1. INTRODUCTION

The AI-Powered Kitchen Ecosystem is a dynamic, web-based platform designed to transform everyday food management into a smart, sustainable experience. Architected with a horizontally layered flow, the system integrates five core components: a User Interface Layer that supports mobile and desktop interactions with camera input; a Frontend Layer built with React for responsive, intuitive user experiences; a Backend Layer powered by Flask and RESTful APIs to handle data routing and logic; an AI Processing Layer that leverages Google Cloud Vision API and machine learning models to analyze food images and generate insights; and a Data Layer anchored by MongoDB and custom recipe databases to store nutritional, behavioral, and historical data. Bidirectional communication, secure connections, and cloud-based scalability ensure seamless collaboration across modules, enabling users to move effortlessly from ingredient recognition to personalized health guidance.

Within this intelligent framework, the AI Shopping Agent serves as the system's smart commerce engine. It enables voice-activated product search, performs real-time price comparisons across multiple retailers, and delivers adaptive recommendations based on user preferences, dietary goals, and ethical filters. Unlike traditional shopping tools, this agent learns from user behavior and integrates with the broader ecosystem—pulling insights from consumed recipes, predicted spoilage, and nutritional needs—to offer context-aware suggestions that evolve over time. It's not just a shopping assistant; it's a personalized, proactive guide that helps users make smarter, healthier, and more sustainable purchasing decisions.

1.1 Background

Let's be honest—online shopping has become the unofficial national sport of the internet [1],[5]. Whether you're hunting for a phone case shaped like a slice of pizza or comparing 47 types of wireless earbuds, the digital marketplace is always open, always tempting, and always just one click away from buyer's remorse .

In theory, it's brilliant: you can buy anything from anywhere, anytime. Pajamas at 2 a.m.? Done. A drone while waiting for your dentist? Why not. But in practice, it's a bit of a circus. With millions of products, confusing filters, and reviews that sound like they were written by Shakespeare's confused cousin, finding the *right* item can feel like solving a mystery novel [11].

And let's not forget the paradox of choice. Too many options can lead to decision fatigue [22] , where you spend three hours choosing a backpack and end up buying socks instead. Or worse buying both and forgetting why you started shopping in the first place.

Most online stores try to help with recommendation systems that say, "You bought a frying pan last week—how about a spatula?" Nice try, algorithm. But these systems often miss the mark, especially when your needs change or your taste evolves. They're like that friend who keeps suggesting horror movies even though you've clearly moved on to rom coms.

1.2 Literature Survey

Researchers have been working hard to tame the chaos of online shopping using smart technologies like Artificial Intelligence (AI) and Natural Language Processing (NLP)[7],[8],[11]. These tools aim to make shopping less of a guessing game and more of a personalized experience—like having a digital butler who knows your style, budget, and irrational fear of glitter.

Traditional recommendation systems rely on past purchases or basic filters [2],[9]. They're helpful, but about as intuitive as a vending machine. They don't understand context, mood, or the fact that sometimes you just want to buy something ridiculous for fun. And most platforms still expect you to type your queries like it's 2005 [12]. Voice search? Still a luxury on many sites [3],[10].

That's where AI and NLP come in. These technologies allow systems to:

- Understand what users *mean*, not just what they *type*
- Learn from user behavior and feedback
- Fetch real-time product info from multiple stores
- Enable voice-based interaction for hands-free shopping
- Convert currencies for global users without needing a calculator

This project proposes an AI Shopping Assistant that's not just smart, it's charming. It listens to your voice, understands your preferences, reads reviews so you don't have to, and even remembers your past searches like a loyal sidekick. It's designed to make shopping feel less like a chore and more like a conversation with your witty, well-informed best friend. In large online stores, this assistant can guide users through chaos, help them find what they need (not just what's trending), and save time while reducing the risk of buying a glow-in-the-dark toilet seat by accident. It's a step toward making online shopping smarter, smoother, and just a little more human.

1.3 Research Problem with Novel Solution

The Problem

Ever tried buying something online and ended up with 47 tabs open, a headache, and still no clue which product is best? Whether it's a rice cooker, gaming mouse, or a pair of noise-canceling headphones, the experience is often chaotic. You're bombarded with options, confusing specs, fake reviews, and prices that change faster than your mood on Monday morning [24],[25].

The core issue: Online shopping platforms don't understand *you*. They don't know your preferences, your budget, or your style. They just throw products at you and hope you click "Add to Cart."

The Novel Solution

Imagine having a personal assistant who listens to your needs—spoken or typed—and finds the perfect product across multiple stores. Not just based on what's popular but based on *you*. That's our AI Shopping Assistant.

What makes it new:

- You can talk to it like a friend: "Find me a nonstick pan under Rs 2000."
- It searches across multiple platforms
- It teaches your preferences over time
- It converts currencies, filters by reviews, and adapts to your feedback
- It's not just a chatbot—it's a smart, evolving shopping companion

No existing system combines voice input, real-time multi-store search, personalization, and ethical filtering in one seamless experience [13],[16]. We're not just building a tool, we're building trust.

1.4 Research Gap

- **Technology Use:**

Most current online shopping systems do not fully combine AI, natural language understanding, live product updates, and learning from user feedback [12],[13] in one complete system.

- **Recommendation Accuracy:**

Many recommendation tools do not check carefully if their suggestions are useful for the user, which can lead to poor or irrelevant product suggestions [24],[25].

- **Guidelines for Ethics:**

If AI systems don't adhere to explicit guidelines, they may occasionally make inaccurate or deceptive recommendations. There is a need for ethical guidelines to make sure users get safe and helpful recommendations [23].

- **Smart Learning:**

There is a need for systems that can keep learning from user feedback, reviews, and shopping trends to improve recommendations over time.

- **Easy Access:**

Many systems are not simple enough for all users to interact with. A friendly interface where users can ask questions in plain language is needed to make shopping easier [17].

- **Complete Approach:**

There is a gap in building a system that combines smart technology, AI, and understanding of user needs to provide accurate, up-to-date, and helpful shopping suggestions for everyone.

1.5 Comparison Table

Feature	Traditional E-Commerce (Amazon, Walmart, eBay)	Popular AI Assistants (Shopmate, Agent One™, Nobi)	Your Assistant
Voice Input	❌ Mostly text-based search	✅ Some support via chat or app	✅ Full voice support
Multi-Store Search	❌ Limited to their own inventory	✅ Some cross-platform tools (e.g., Nobi)	✅ Real-time across multiple stores
Real-Time Price Comparison	❌ Manual comparison required	✅ Limited, often delayed or static	✅ Live price updates across platforms
Personalized Recommendations	✅ Based on purchase history	✅ Context-aware, behavior-based	✅ Adaptive learning with feedback loop
Currency Conversion	❌ Often region-locked pricing	❌ Rarely built-in	✅ Automatic, global-friendly conversion
Ethical Filtering	❌ No transparency or bias control	❌ Not clearly defined	✅ Built-in ethical rules and review analysis

Table 1 -- Comparison Table

1.6 Identification of Research Area

1.6.1 Research Domain

This project isn't just about building software—it's about creating a shopping experience that feels smart, intuitive, and even a little magical. To make that happen, we're drawing from five key areas of tech:

- **Artificial Intelligence (AI):** This is the assistant's brain. It learns what users like, what they don't, and makes smart suggestions that actually make sense.
- **Natural Language Processing (NLP):** This is how the assistant understands everyday language—whether you say "cheap rice cooker" or "something quiet for smoothies," it gets the idea.
- **Human-Computer Interaction (HCI):** We want users to feel like they're chatting with a helpful friend, not wrestling with a robot. HCI helps us design that friendly vibe.
- **E-commerce Systems:** The assistant connects with real online stores to grab product info, prices, and reviews—so users get real-time, relevant results.
- **Voice Technology:** Because sometimes you just want to say what you need and let the assistant do the rest—hands-free, hassle-free.

1.6.2 Tools & Technologies

Layer	Tools & Technologies
Core Development	Python – the foundation of everything
Voice Input	SpeechRecognition, PyAudio, Web Speech API – so users can talk instead of type
NLP & Intent Recognition	spaCy, TextBlob, TensorFlow – to understand what users *really* mean
Web Scraping	Selenium, BeautifulSoup – to fetch live product data from stores
Recommendation Engine	TensorFlow, Scikit-learn – to suggest products users will actually love
Review Analysis	OpenAI API – to summarize reviews and highlight what matters
Frontend Interface	React.js – for a clean, responsive, and friendly user experience
Backend Services	Flask, FastAPI – to keep everything connected and running smoothly
Data Storage	MongoDB, Firebase – to remember user preferences and search history
Currency Conversion	forex-python – so prices make sense no matter where you shop from
Version Control	GitHub, GitLab – to keep the code organized and collaborative
Deployment & Scalability	Cloud Services – so the assistant can grow with its users and never slow down

Table 2 -- Tools & Technologies

2. OBJECTIVES

2.1 Main Objectives

- **Create a Smart Kitchen Shopping Helper:** Build an AI assistant that understands what users want when buying kitchen items online and guides them to the best products.

- **Give Helpful Suggestions:** Provide recommendations and tips that match the user's needs, budget, and cooking style.

- **Make Shopping Easy and Enjoyable:** Offer clear, friendly, and useful advice so buying kitchen products is simple and stress-free.

- **Adapt to User Preferences:** Notice changes in the user's needs or interests and update recommendations accordingly.

- **Talk in a Natural Way:** Keep the conversation smooth, clear, and on topic, even if the user changes what they are looking for.

2.2 Specific Objectives

- **Understand Users Better:** Learn from what the user types, says, or searches to know their kitchen product preferences.

- **Give the Right Product Matches:** Use user preferences and needs to suggest the best kitchen products in a friendly way.

- **Improve Over Time:** Learn from feedback and past shopping history to make better suggestions in the future.

- **Protect User Privacy:** Keep all personal and shopping data safe and private while still giving personalized help.

2. METHODOLOGY

3.1 Overall System Architecture

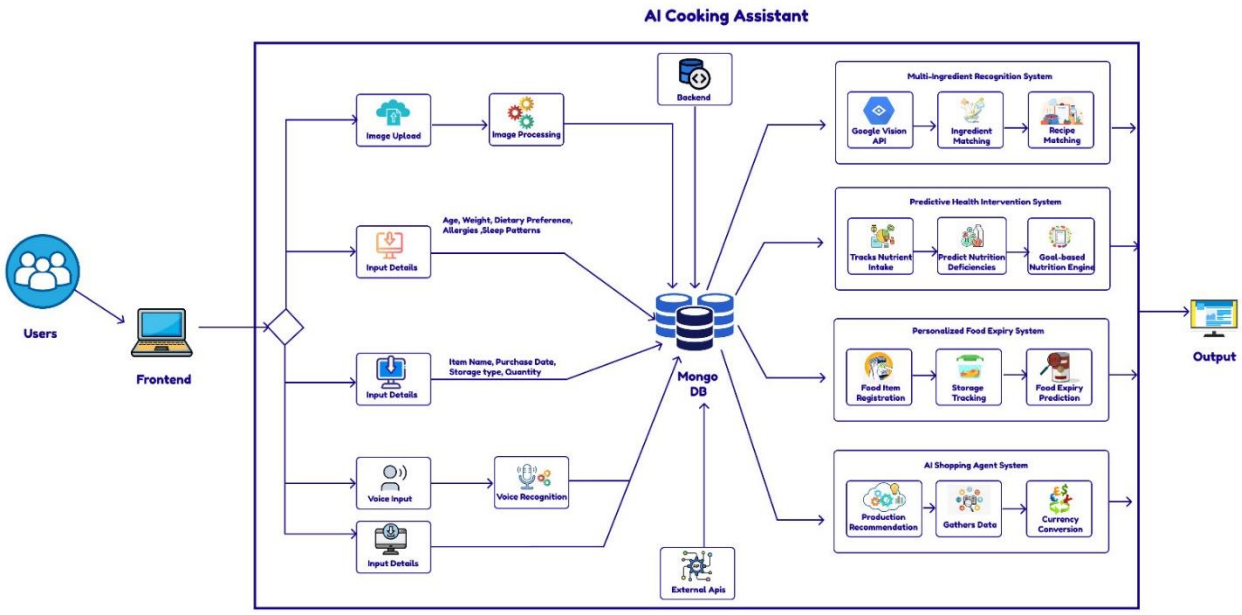

Figure 1 -- Overall System Architecture

3.2 System Architecture

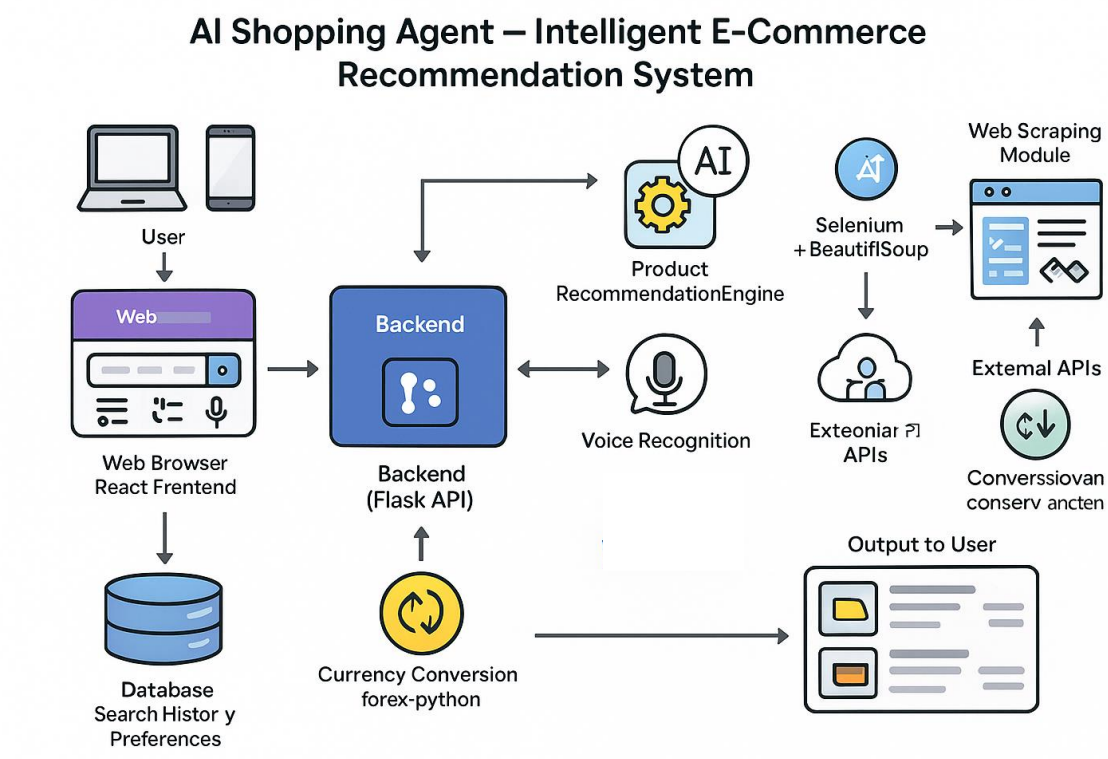

Figure 2 -- System Architecture

3.2 System Architecture Overview – AI Shopping Assistant

The system architecture of the AI Shopping Assistant is designed to deliver smart, personalized shopping experiences. It includes[14],[15]:

- **Frontend (React):** Where users interact—searching, speaking, and viewing product suggestions.
- **Backend (Flask API):** The central hub that connects all modules and handles logic.
- **Voice Recognition:** Converts spoken input into text for hands-free shopping.
- **Recommendation Engine:** Uses AI and NLP to suggest relevant products.
- **Web Scraper:** Collects real-time data from e-commerce sites like eBay and Walmart.
- **Currency Converter:** Adjusts prices to the user's local currency.
- **Database:** Stores user preferences, history, and feedback.
- **External APIs:** Provide extra product details and analytics.

Together, these components create a seamless flow—from user input to intelligent product recommendations—making online shopping smarter and more human-centered.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

- User input Understanding: Receive and comprehend the information entered or voiced by the user, to determine their purchasing needs.
- Dynamic Product Search: Quick search across different online stores for a particular item.
- Recommendation: Recommend to the user items that correspond to their taste, budget, and past search history.
- Smooth Search and Filters: Easily searchable and filterable to allow users quick access to desired products.
- Secure User Data: Ensure the safety of all personal information of every user while they make a purchase.
- User-Friendly: Keep the purchase process simple and clear for all.
- Feedback and Ratings: Allow the users to rate the products and write reviews to enhance the system.
- Work with Other Systems: Connect with e-commerce sites, currency tools, and AI services.

4.2 Non-functional requirements

- Fast speed of displaying search results and recommendations This website provides the fastest speed of showing search results and recommendations.
- Accurate and personal recommendations The website provides the most accurate and personal recommendations.
- Easy to use, simple, clean design User-friendly design. It is easy to use and has a simple and clean design.
- Always available All time working without any break Secure All data will be secure from hackers or misuse Work everywhere Good performance on all devices and browsers Keep users interested by incorporating alerts, chat, or updates

4.3 Expected Test Cases

Test Case ID	Description	Input	Expected Output
TC-01	Search for a kitchen product by text	Search "Non-stick Frying Pan"	Shows list of non-stick frying pans from different stores
TC-02	Search for a kitchen product by voice	Say "Electric Kettle"	Shows list of electric kettles from different stores
TC-03	Filter kitchen products	Apply filter: Price below $50	Only kitchen products under $50 are shown
TC-04	Sort kitchen products	Sort by "Customer Rating: High to Low"	Products listed with highest-rated items first
TC-05	Show personalized kitchen recommendations	User likes baking items	Suggests baking tools, cake mixers, and ovens
TC-06	Compare prices of a kitchen item	Select "Microwave Oven"	Shows prices from all connected stores for the same model
TC-07	Change currency for kitchen products	Switch USD → GBP	Prices are updated in British pounds
TC-08	Save kitchen product search history	Search "Blender"	Search is saved in history list
TC-09	Delete kitchen product search history	Delete "Blender" from history	Item is removed from search history
TC-10	Give kitchen product feedback	Rate "Stainless Steel Knife Set" 5 stars	Feedback is saved successfully
TC-11	Mobile access to kitchen shopping	Open on mobile browser	Kitchen products search works and looks good on mobile
TC-12	Speed check for kitchen searches	Search "Pressure Cooker"	Results show within 3 seconds
TC-13	No results found for kitchen item	Search "XYZ Kitchen Tool"	Shows "No products found" message
TC-14	No internet while searching	Turn off internet and search "Mixer Grinder"	Shows "Check your internet connection" message

Table 3 -- Expected Test Cases

5. GANTT CHART

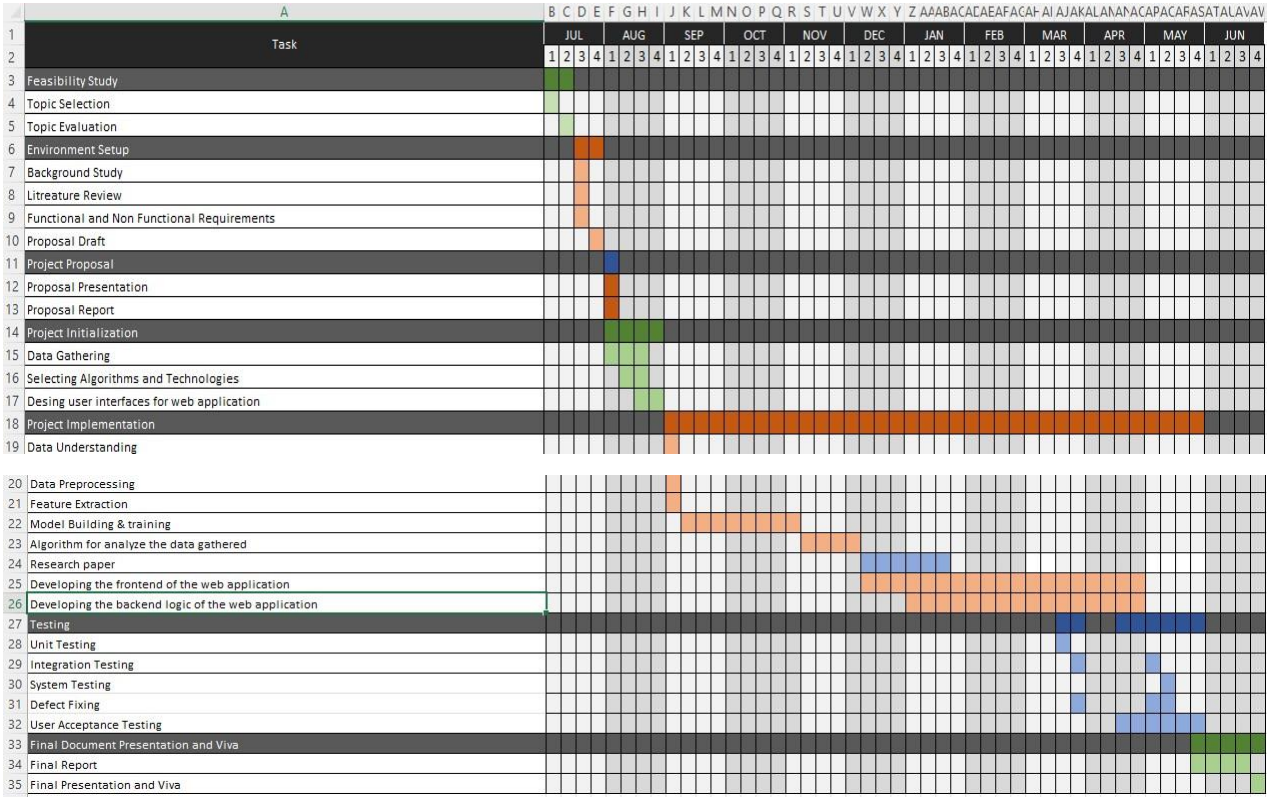

Figure 3 -- Gannt Chart

6. WORK BREAKDOWN STRUCTURE

Phase	Tasks	Timeline
Month 1 – Initiation	Topic selection, literature review, define scope, finalize objectives	Week 1–4
Month 2 – Proposal	Write and submit proposals, prepare proposal presentation	Week 5–8
Month 3 – Design	System architecture, UI mockups, user flow diagrams	Week 9–12
Month 4 – Progress Report 1	Submit first progress report, present initial prototype & design decisions	Week 13–16
Month 5–6 – Development Phase 1	Build backend/frontend, integrate basic NLP & voice modules	Week 17–24
Month 7 – Progress Report 2	Submit second progress report, present updated system & early results	Week 25–28
Month 8–9 – Development Phase 2	Refine features, improve recommendation engine, enhance UX	Week 29–36
Month 10 – Testing & Evaluation	Conduct user testing, analyze feedback, iterate based on results	Week 37–40
Month 11 – Finalization	Polish UI, optimize performance, complete documentation	Week 41–44
Month 12 – Final Presentation & Submission	Prepare final report, presentation slides, viva prep, submit project	Week 45–48

Table 4 -- work breakdown structure

7. Ability of Commercialization

This isn't just a cool project, it's a product with real market potential.

Real-World Applications

- E-commerce platforms to boost conversion rates
- Retailers to offer personalized shopping experiences
- Mobile apps for smart shopping
- Voice-enabled kiosks in stores

Revenue Models

- Affiliate commissions from purchases
- Premium subscription for advanced features
- B2B licensing to online retailers

Scalability

- Expand to fashion, electronics, groceries, travel bookings—you name it

- Localize for different languages and currencies
- Integrate with smart home devices (e.g., “Hey Assistant, order me coffee pods”)

8. Budget and Budget Justification (if any)

For this research component, we do not aim to commercialize our individual hardware or IoT modules separately. Instead, the whole research will be implemented and tested as an **AI-Powered Kitchen Ecosystem for Food Waste Reduction and Nutrition Optimization**. The following budget justification outlines the anticipated costs for the development, testing, and validation of the system.

Description	Cost (LKR)
Electricity	5,000.00
Internet	5,000.00
Communication	2,000.00
Transportation	3,000.00
Paper Publishing	15,000.00
Backend Hosting	12,000.00
Software & API Services	10,000.00
Prototype Materials (Sensors, Basic Devices)*	8,000.00
Total	**60,000.00**

table 5 -- Budget and Budget Justification

8.1 Budget Justification

1. **Electricity (LKR 5,000):** Covers power consumption during system development, AI model training, and testing.
2. **Internet (LKR 5,000):** High-speed internet for research, cloud services, and API access.
3. **Communication (LKR 2,000):** Costs for phone calls, mobile data, and collaboration platform subscriptions to ensure seamless coordination among the research team
4. **Transportation (LKR 3,000):** Travel for research meetings, presentations, or data collection if needed.
5. **Paper Publishing (LKR 15,000):** Costs for conference or journal publication of the research findings.
6. **Backend Hosting (LKR 12,000):** Cloud services for hosting the system and ensuring scalable access.
7. **Software Purchasing (LKR 10,000):** Licenses for necessary software, IDEs, databases, and AI libraries not available for free.

8. **Prototype Materials (LKR 8,000):** Funds for basic hardware components (e.g., sensors, microcontrollers) to build and test a physical proof-of-concept.

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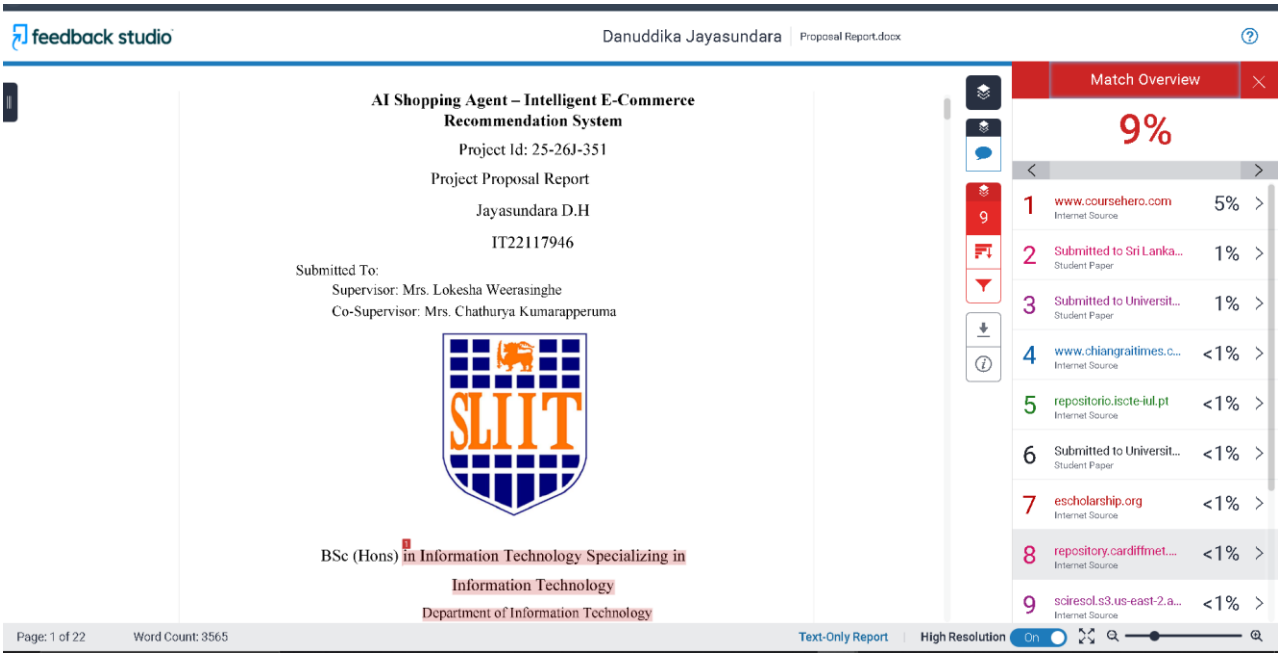
10. APPENDICES

Appendix - A: Plagiarism Report



Appendix 1 -- Plagiarism Report